

Exploratory Analysis of a 50,000-Patient Synthetic Diabetes Risk Dataset: Strong Univariate Separability with Near-Absent Multivariate Structure

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Abstract—This work presents an exploratory analysis of a publicly available synthetic dataset of diabetes risk, with the goal of characterising which of its predictors carry discriminative information before any predictive model is considered. The dataset comprises $N = 50,000$ records, $D = 22$ predictors and $L = 3$ risk classes, with a strongly imbalanced class distribution in which the *Low* class holds only 0.94% of the observations. We combine unconditional and class-conditional univariate statistics, pairwise Pearson correlation and a principal component analysis (PCA) implemented from first principles, without recourse to pre-made PCA routines. Two glycemic markers, fasting blood sugar and HbA1c, separate the risk classes almost by themselves, yet the first two principal components account for only 15.2% of the total variance. Three independent lines of evidence, an excess kurtosis of -1.2 in 19 of the 20 quantitative predictors, a median absolute pairwise correlation of 0.003, and an eigenvalue spectrum flat at unity, converge on the same conclusion: the predictors were drawn independently from bounded uniform distributions, and the only structural relationship the data contain is the algebraic identity that defines the body mass index.

Index Terms—exploratory data analysis, principal component analysis, synthetic data, diabetes risk, class separability

I. INTRODUCTION

Diabetes mellitus is one of the largest public health burdens worldwide [1]. In 2024, 589 million adults aged 20 to 79 years, one in every nine, were living with the condition, and this figure is projected to reach 853 million by 2050, a 45% rise against a projected population growth of 25%. A further 252 million adults are estimated to be unaware that they have it. The disease accounted for 3.4 million deaths and for at least USD 1 trillion in health expenditure in that single year, which is what makes early detection consequential: it opens a window for treatment and lifestyle intervention before microvascular and cardiovascular complications become established.

Fitting a classifier without a prior exploratory analysis makes it impossible to tell whether the performance obtained reflects genuine clinical patterns or artefacts of the data. Descriptive statistics and linear projections expose, beforehand,

redundancy among predictors, disparities of scale, outliers and the degree of overlap between the classes. Without that map of the underlying structure, neither the success nor the failure of a model can be diagnosed.

The use of synthetic datasets in health care has grown as a way of circumventing strict patient privacy constraints, of compensating for the scarcity of records and of enabling educational applications [2]. That convenience, however, carries the risk that the generating process fails to reproduce the complex physiological interdependencies present in real clinical populations. When multivariate relationships are not preserved, algorithms trained on such data tend to learn mathematical artefacts rather than legitimate biological mechanisms. It is in this context that the dataset analysed here belongs, its origin being explicitly synthetic [3].

This work presents an exploratory and multivariate characterisation of a synthetic dataset of diabetes risk, assessing its statistical consistency and its clinical fidelity before any modelling step. We investigate the dependencies among predictors and determine to what extent the risk classes are structurally separable, both in the original predictor space and under projections of maximum variance. The remainder of this paper is organised as follows. Section II describes the dataset and the analytical methods. Section III reports and discusses the results. Section IV concludes.

II. METHODS

A. Dataset

The dataset was obtained from Kaggle [3] and is explicitly described by its authors as a large-scale *synthetic* healthcare dataset. The raw file holds 41 columns, from which we discard `Patient_ID`, a row identifier carrying no physiological information, and `Diabetes_Risk_Score`, the continuous score from which the categorical label is derived. This leaves $N = 50,000$ observations and $D = 22$ predictors, of which 20 are quantitative and two (`Country` and `Gender`) are

categorical. The class label `Diabetes_Risk` takes $L = 3$ values.

Table I reports the class distribution. The disparity between the 36,593 observations of class *High* and the 470 of class *Low*, a ratio of 78 to 1 and a mere 0.94% of the sample, imposes structural limits on the analysis. It renders the class-conditional estimates of the minority class unstable, and it would compromise the evaluation of any future classifier, since a naive model that constantly predicts *High* would already reach 73% accuracy.

TABLE I
CLASS DISTRIBUTION OF `DIABETES_RISK`

Class	N_l	Share
High	36,593	73.19%
Moderate	12,937	25.87%
Low	470	0.94%
Total	50,000	100.00%

Table II summarises the missing values, which affect 12 of the 20 quantitative predictors. Four of them are missing *exactly* one thousand values each, and those four are precisely the complete lipid panel: total cholesterol, HDL, LDL and triglycerides. A fifth column outside the quantitative set, medication adherence, shows the same round count. Natural record loss does not produce an identical round figure across five variables, so the absences were evidently injected by the generator. They were not injected by examination, however: no observation lacks all four lipid measurements at once, and the pairwise overlap between them ranges from 12 to 21 rows, which is what independent sampling of 2% of the records would yield. In clinical practice a lipid panel is a single blood draw and its four results go missing together; here a fixed quota of one thousand absences was drawn column by column, ignoring that dependency.

TABLE II
MISSING VALUES (ONLY AFFECTED PREDICTORS SHOWN)

Predictor	Missing
Weight kg	2,962
Height cm	2,945
Sleep Hours	2,017
Exercise Hours Per Week	1,992
HbA1c	1,974
Daily Walking Minutes	1,964
LDL	1,000
HDL	1,000
Total Cholesterol	1,000
Triglycerides	1,000
Blood Glucose	959
Age	497

B. Pre-processing

Standardising the variables to zero mean and unit variance before the principal component analysis is mandatory, since the raw predictors operate on widely disparate scales (mg/dL,

kg, cm, years); without that correction the variables of larger magnitude would dominate the covariance matrix and the analysis would describe the units of measurement rather than the intrinsic structure of the data. Furthermore, because the covariance matrix requires complete records, the rows holding missing values were removed. This exclusion reduced the set from 50,000 to 33,689 observations, that is, 32.6% of the data discarded, and it establishes a methodological limitation: the multivariate analysis describes a complete subset and not the entire cohort. The univariate and bivariate statistics, in contrast, use every observation available for each predictor.

C. Analytical methods

For each predictor d we compute the mean μ_d , the standard deviation σ_d and the skewness γ_d ,

$$\gamma_d = \frac{1}{N} \sum_{n=1}^N \left(\frac{x_{n,d} - \mu_d}{\sigma_d} \right)^3, \quad (1)$$

which quantifies the imbalance between the tails of a distribution and vanishes for any density symmetric about its mean, together with the excess kurtosis, the analogous fourth-order moment minus three, and their class-conditional counterparts $\mu_{d|l}$, $\sigma_{d|l}$ and $\gamma_{d|l}$ over the N_l observations of each class l . Skewness vanishes for any symmetric distribution and so cannot by itself distinguish a Gaussian from a uniform density, whereas the excess kurtosis separates them cleanly: it is 0 for the Gaussian and $-6/5$ for the uniform. We therefore report both.

Linear dependence between pairs of predictors is quantified by the Pearson correlation coefficient

$$\rho_{d_i, d_j} = \frac{\mathbb{E} [(X_{d_i} - \mu_{d_i})(X_{d_j} - \mu_{d_j})]}{\sigma_{d_i} \sigma_{d_j}}, \quad (2)$$

which captures linear association only. A vanishing ρ does not imply independence in general, so we treat the correlation matrix as one piece of evidence among three rather than as a conclusion.

To rank predictors by their ability to separate classes on their own, we use the ratio between the range of the class-conditional means and the unconditional standard deviation,

$$\delta_d = \frac{\max_l \mu_{d|l} - \min_l \mu_{d|l}}{\sigma_d}, \quad (3)$$

so that δ_d expresses the class separation in units of the predictor's own spread and is comparable across predictors measured in different physical units.

Finally, the principal component analysis was implemented from scratch, without pre-made PCA functions or libraries, following the standard chain [4]: standardisation, estimation of the covariance matrix $\mathbf{S}$, eigendecomposition $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, and projection of the observations onto the two eigenvectors of largest eigenvalue. The only library routine used is the symmetric eigensolver.

III. RESULTS AND DISCUSSION

A. Unconditional univariate analysis

Table III reports μ_d , σ_d , γ_d , the excess kurtosis and the observed range of the 20 quantitative predictors, and Fig. 1 shows the corresponding histograms.

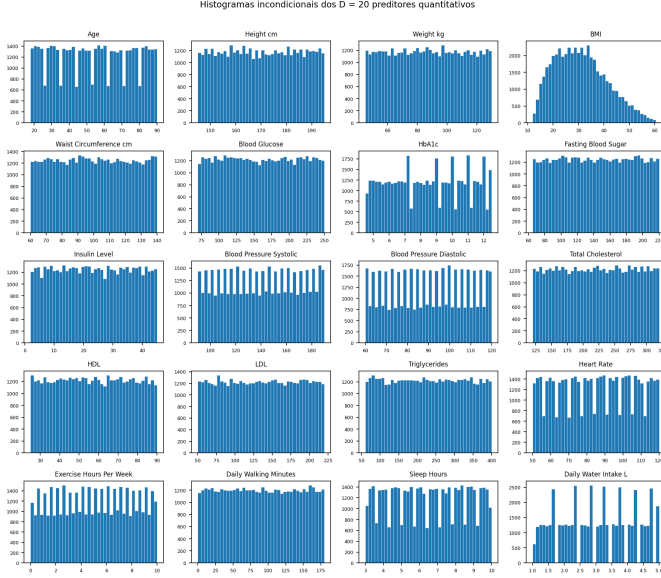


Fig. 1. Unconditional histograms of the 20 quantitative predictors. All but the body mass index are visibly flat-topped.

Nearly every predictor shows a practically null skewness, ranging from -0.015 to $+0.008$, with the notable exception of BMI ($\gamma = 0.432$). Such symmetry, however, does not imply normality, since a uniform density is strictly symmetric as well; the decisive quantity is the kurtosis. Nineteen of the twenty predictors have an excess kurtosis between -1.21 and -1.19 , matching the theoretical value of $-6/5$ expected for a uniform distribution. This is corroborated by the histograms of Fig. 1, which are visibly flat and truncated at round limits chosen for convenience rather than at physiological ones: age from 18 to 90 years, heart rate from 50 to 120 bpm, triglycerides from 60 to 400 mg/dL and sleep from 3 to 10 hours. The reading these figures impose is that the predictors do not reflect measurements taken from real patients, but were drawn from predefined uniform distributions. This is the first of three analytical signals that will converge in Section III-D.

Body mass index escapes this otherwise general artificial pattern, showing a positive skewness and a kurtosis of -0.45 against the -1.2 of the remaining variables. The divergence arises because BMI was not drawn by the generator but computed algebraically as the quotient $\text{weight}/\text{height}^2$, and the ratio of two independent uniform variables is not itself uniform. The exception therefore confirms that reading rather than weakening it: BMI is the one predictor the generator did not draw.

B. Class-conditional analysis and discriminative power

Table IV reports the class-conditional means and ranks the predictors by the discriminative power of (3); Fig. 2 shows the class-conditional histograms and Fig. 3 the box-plots of the four strongest predictors.

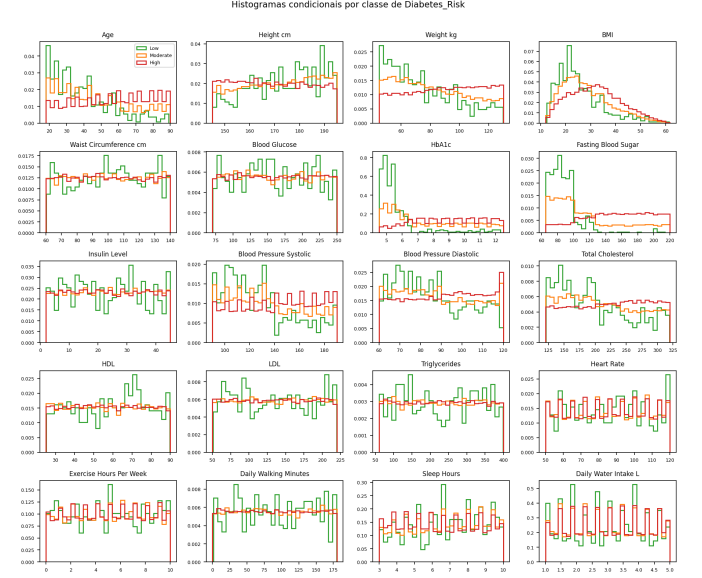


Fig. 2. Class-conditional histograms of the 20 quantitative predictors, normalised to unit area.

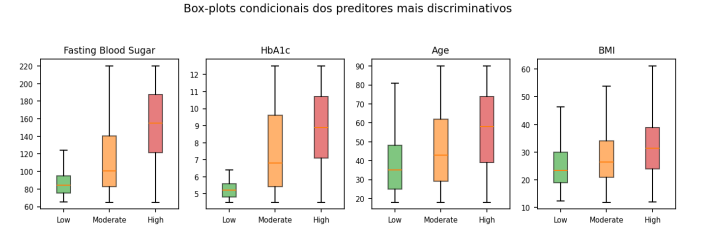


Fig. 3. Class-conditional box-plots of the four most discriminative predictors.

The answer to whether any predictor separates the classes on its own is affirmative, and two markers stand out: fasting blood sugar ($\delta = 1.44$) and HbA1c ($\delta = 1.38$). The discriminative capacity then falls substantially, to age ($\delta = 0.84$), BMI ($\delta = 0.61$) and weight ($\delta = 0.57$). The remaining fifteen predictors all stay below $\delta = 0.5$, and ten of them below $\delta = 0.1$, a range in which the separation between groups is indistinguishable from noise.

From a clinical standpoint, the predictors with the greatest separating power display values and progressions that make full physiological sense [5]. HbA1c rises from 5.67% in the low-risk class to 8.86% in the high-risk one, which is exactly the transition from the normal to the diabetic range, while fasting blood sugar goes from 88.4 to 152.8 mg/dL, comfortably crossing the diagnostic threshold of 126 mg/dL. Age advances from 38.8 to 56.5 years and BMI evolves from 25.6 to 31.9, a clear progression from overweight to obesity.

TABLE III
UNCONDITIONAL UNIVARIATE STATISTICS OF THE 20 QUANTITATIVE PREDICTORS. THE EXCESS KURTOSIS OF A UNIFORM DISTRIBUTION IS $-6/5$.

Predictor	μ_d	σ_d	γ_d	Excess kurtosis	Range
Age	53.87	21.13	+0.008	-1.206	18.0–90.0
Height cm	170.02	14.46	+0.000	-1.207	145.0–195.0
Weight kg	87.46	24.50	+0.004	-1.191	45.0–130.0
BMI	30.92	10.29	+0.432	-0.453	11.9–61.7
Waist Circumference cm	99.96	23.09	+0.010	-1.191	60.0–140.0
Blood Glucose	159.91	52.04	+0.006	-1.212	70.0–250.0
HbA1c	8.49	2.31	+0.002	-1.204	4.5–12.5
Fasting Blood Sugar	142.61	44.68	-0.001	-1.200	65.0–220.0
Insulin Level	23.48	12.40	+0.006	-1.200	2.0–45.0
Blood Pressure Systolic	140.17	29.17	-0.004	-1.201	90.0–190.0
Blood Pressure Diastolic	90.04	17.58	-0.010	-1.193	60.0–120.0
Total Cholesterol	220.17	57.79	-0.005	-1.203	120.0–320.0
HDL	57.37	18.72	+0.004	-1.195	25.0–90.0
LDL	135.01	49.14	-0.004	-1.205	50.0–220.0
Triglycerides	229.54	98.23	-0.000	-1.200	60.0–400.0
Heart Rate	85.12	20.39	-0.015	-1.189	50.0–120.0
Exercise Hours Per Week	5.02	2.88	-0.009	-1.192	0.0–10.0
Daily Walking Minutes	89.93	52.03	+0.005	-1.206	0.0–180.0
Sleep Hours	6.50	2.03	-0.005	-1.210	3.0–10.0
Daily Water Intake L	3.00	1.15	+0.003	-1.189	1.0–5.0

TABLE IV
CLASS-CONDITIONAL MEANS AND DISCRIMINATIVE POWER δ_d ,
PREDICTORS SORTED BY δ_d

Predictor	$\mu_{d Low}$	$\mu_{d Mod}$	$\mu_{d High}$	δ_d
Fasting Blood Sugar	88.44	115.81	152.78	1.44
HbA1c	5.67	7.54	8.86	1.38
Age	38.81	46.89	56.53	0.84
BMI	25.60	28.31	31.92	0.61
Weight kg	75.68	81.77	89.61	0.57
Blood Pressure Systolic	127.98	134.73	142.25	0.49
Total Cholesterol	200.41	211.82	223.37	0.40
Blood Pressure Diastolic	85.55	88.06	90.80	0.30
Height cm	173.66	171.76	169.35	0.30
Sleep Hours	6.78	6.66	6.44	0.17
HDL	58.68	57.25	57.40	0.076
Daily Water Intake L	2.94	2.98	3.01	0.059
Insulin Level	23.05	23.61	23.44	0.045
Waist Circumference cm	100.85	100.08	99.91	0.041
Daily Walking Minutes	91.20	90.24	89.81	0.027
Blood Glucose	159.01	159.46	160.07	0.020
Heart Rate	85.36	85.10	85.13	0.013
LDL	135.04	135.35	134.90	0.009
Exercise Hours Per Week	5.03	5.02	5.02	0.006
Triglycerides	229.47	229.47	229.57	0.001

Under this strict reading of isolated means, the dominant predictors are the clinically correct ones.

Three observations qualify this apparent clinical fidelity. First, casual blood glucose has a practically null discriminative power ($\delta = 0.020$) and behaves as mere noise, whereas fasting blood sugar dominates the separation ($\delta = 1.44$). Since both measure the same physiological substrate and move together in a real patient, the generator evidently derived the labels from one and ignored the other.

Second, waist circumference, although an established marker of metabolic and cardiovascular risk, does not separate the classes in this synthetic set ($\delta = 0.041$). In further contradiction with physiology, its mean value decreases slightly as

the diabetic risk increases, from 100.85 to 99.91 cm.

Third, mean height decreases from 173.7 to 169.4 cm as risk rises, with $\delta = 0.30$, which has no medical basis whatsoever. This distortion is a strictly algebraic consequence of BMI driving the assignment of the labels: at equal weight, shorter individuals have a mathematically larger BMI, so height inherits an inverted correlation with the disease that it would never hold in a real population.

C. Bivariate analysis

Fig. 4 shows the correlation matrix and Fig. 5 the pairwise scatter plots, coloured by class.

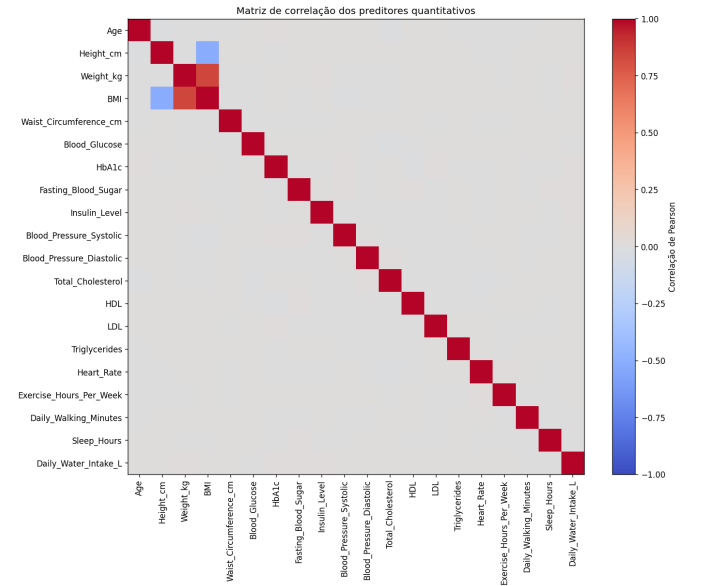


Fig. 4. Pairwise Pearson correlation matrix of the 20 quantitative predictors.

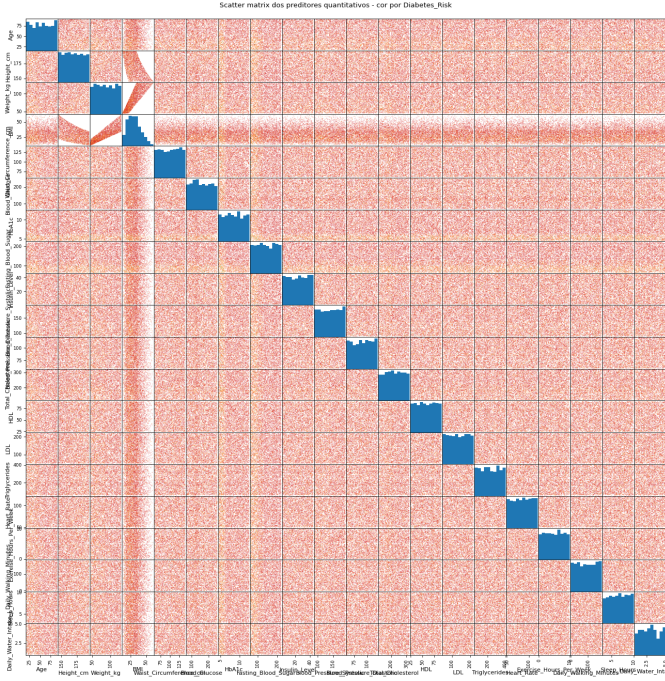


Fig. 5. Pairwise scatter plots of the quantitative predictors on a random subsample of 3,000 observations, coloured by risk class.

The correlation matrix approaches an identity matrix closely, with a median absolute off-diagonal correlation of 0.003. Only two pairs depart from this pattern of total independence: BMI with weight ($\rho = +0.843$) and BMI with height ($\rho = -0.516$). These two coefficients, however, reflect nothing but the mathematical definition of the index itself, and are not an empirical or biological finding. Consistently with uniform and independent predictors, the point clouds of Fig. 5 form rectangles devoid of latent structure, and no pair of variables separates the classes visually. This absence of linear dependence is the second signal.

The least plausible finding of the correlation matrix is that the only two predictors with relevant discriminative power are statistically uncorrelated with each other ($\rho = -0.005$). In a real patient HbA1c and fasting blood sugar are strongly correlated, since the former integrates the average glycaemia of the preceding weeks. Here the disease label depends on both variables along mathematically independent paths, which is physiologically impossible and is explained only by the rules of the generating process, and it is the sharpest illustration of that second signal.

D. Principal component analysis

Table V lists the eigenvalue spectrum and Fig. 6 shows the projection onto the first two components.

After standardisation the eigenvalues sum to 20, so a set of mutually uncorrelated predictors would yield twenty eigenvalues equal to 1. That is very nearly what is observed: components PC2 to PC19 lie between 0.956 and 1.037, confined to a band of less than 5% around unity. Such a

TABLE V
EIGENVALUE SPECTRUM OF THE STANDARDISED COVARIANCE MATRIX
(20 COMPONENTS, 33,689 COMPLETE OBSERVATIONS)

Component	λ	Explained	Cumulative
PC1	1.9943	9.97%	9.97%
PC2	1.0371	5.19%	15.16%
PC3	1.0327	5.16%	20.32%
PC4–PC19	0.9564–1.0292	—	—
PC20	0.0123	0.06%	100.00%

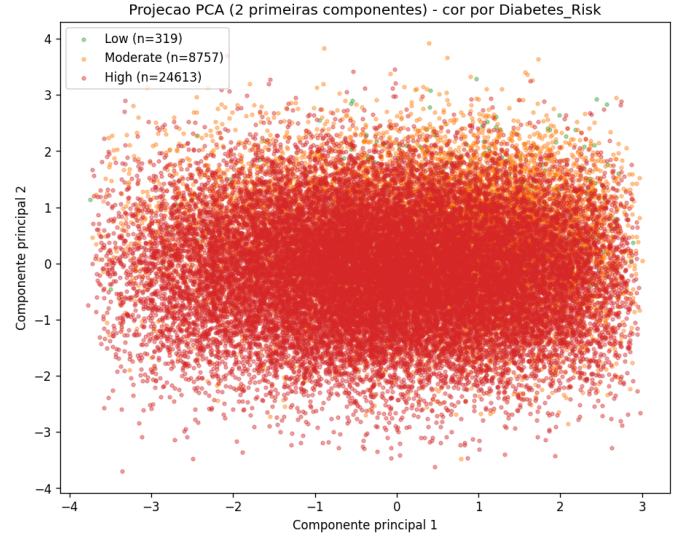


Fig. 6. Observations projected onto the first two principal components, coloured by risk class.

flat spectrum is the classical signature of an absence of linear structure. It constitutes the third independent signal, produced by a third statistical method, pointing at the same fact that the kurtosis analysis and the correlation matrix had already indicated.

The only departures from the flatness of the spectrum are the first component ($\lambda = 1.994$) and the last one ($\lambda = 0.0123$). The first concentrates its loadings almost exclusively on BMI (-0.706), weight (-0.602) and height ($+0.372$), everything else staying below 0.02 in absolute value, while the last mirrors exactly the same triple with BMI $+0.708$, weight -0.603 and height $+0.368$. The reading is that the analysis isolated, unprompted, the algebraic identity by which BMI is strictly derived from weight and height. The largest variance of the set therefore lies in the plane spanned by these three variables, and the near-null eigenvalue measures the almost deterministic dependence among them, almost and not exactly because the relation is non-linear and the method captures only its best linear approximation. It is the single structural relation present in the data.

Answering the three questions directly, the classes are not well separated in the projection, and are in fact separated *worse* than in the univariate analysis of HbA1c or fasting blood sugar. Second, no linear decision boundary is visible

that would distinguish the risk groups in the projected space. Finally, *Moderate* overlaps *High* heavily, while *Low* becomes practically invisible, both because of its small sample size, only 470 points, and because it occupies the very same region as the other two.

The apparent contradiction between the strong signal found in the univariate analysis and the absence of separation under PCA dissolves once it is understood that the two measure distinct properties. The first two components explain only 15.2% of the variance because the spectrum is flat and no dominant direction exists to concentrate it. The projection fails to separate the classes for a different reason: the method maximises variance, not separability, and when the predictors are almost all independent there is no high-variance direction in which the class information happens to lie. Moreover, the single high-variance direction the set does possess, that of BMI, is merely an algebraic relation with no discriminative content of its own. PCA did not fail; it correctly reported that there is no multivariate structure to report.

IV. CONCLUSION

The exploratory analysis produced three independent pieces of evidence that converge on the same fact: the uniform kurtosis of nineteen of the twenty predictors, the median of 0.003 among the off-diagonal correlations, and the flat eigenvalue spectrum. Together they indicate that the predictors were sampled from independent uniform distributions over round numerical intervals, and that the diagnostic label was assigned afterwards as a function of a restricted group of variables, chiefly fasting blood sugar, HbA1c, age and BMI. Consequently the only structural interdependence present in this dataset is strictly algebraic, with no grounding in real physiology.

In practical terms, a synthetic dataset with these characteristics serves perfectly well to exercise the methodological flow of data analysis, which is the scope and the objective of this work. It is, on the other hand, strictly inadequate as a basis for clinical conclusions or for realistically testing multivariate methods, precisely because the covariance those algorithms exist to exploit was suppressed at generation time. A classifier trained here would reach a very high apparent performance while relying on two predictors alone, and that behaviour would not transfer to real medical records. Recording and documenting this structural artificiality is, in itself, the most robust result this exploratory analysis delivers.

USE OF AI TOOLS

AI assistance (Anthropic Claude) was used in this work for code debugging, for verification queries over the dataset, for structuring the text and for checking the references. The prompts, the responses and the use made of each are recorded in `USO_DE_IA.md` in the project repository.

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